

Student Club Event Ticketing & Budget Portal

Course: Software Engineering

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Introduction

Background

Student clubs frequently organize campus events that require proposal approval, budget management, ticket generation, and participant registration. Traditional paper-based or manual processes are time-consuming and make it difficult to track approvals, payments, and attendance. A digital portal simplifies these activities by providing a centralized platform for event management.

Purpose

The purpose of the Student Club Event Ticketing & Budget Portal is to provide an online system that allows club leaders to submit event proposals, campus administrators to review and approve budgets, and students to purchase event tickets with QR codes. The system improves efficiency, transparency, and record management throughout the event lifecycle.

Scope

The system supports the following major functions:

- Event proposal submission
- Budget approval and rejection
- QR-code ticket generation
- Event ticket booking
- Viewing and downloading QR tickets
- Monitoring event registrations
- User authentication and authorization

These functions are derived from the functional requirements in the report

System Overview

The Student Club Event Ticketing & Budget Portal is a web-based application designed to streamline the management of student club events. The system provides separate functions for Club Leads, Campus Admins, and Students through role-based access control.

The Club Lead creates event proposals, requests budgets, generates QR-code tickets after approval, and monitors event registrations. The Campus Admin reviews submitted proposals and approves or rejects budget requests. Students browse approved events, purchase tickets, and download their QR-code tickets after successful booking. These roles and their responsibilities match the system actors defined in the report.

Functional Requirements

And

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow the Club Lead to create and submit an event proposal containing event name, date, venue, description, expected participants, and proposed budget.	High	The Club Lead can enter all mandatory event details and submit the proposal. The system validates the required fields and generates a unique event ID.	Enables Club Leads to formally submit events for campus approval.
FR-002	Functional	The system shall allow the Campus Admin to review submitted event proposals and approve or reject the proposed budget.	High	The Campus Admin can view event and budget details, approve or reject the proposal, and provide a reason when rejecting. The decision is recorded and visible to the Club Lead.	Provides controlled review and authorization of club events and expenditure.
FR-003	Functional	The system shall allow the Club Lead to generate event tickets with unique QR codes for an event after the event has been approved.	High	The Club Lead can generate tickets only for an approved event. Each generated ticket has a unique ticket ID and QR code, and ticket availability is recorded by the system.	Enables authorized events to issue digitally verifiable tickets.

FR-004	Functional	The system shall allow the Student to purchase tickets for approved events and receive a unique QR-code ticket after successful payment.	High	The Student can view approved events, select an event and ticket quantity, verify availability, complete payment, and receive a unique QR-code ticket. The purchase is recorded and available ticket count is updated only after successful payment.	Enables students to digitally purchase tickets and receive verifiable event access.
FR-005	Functional	The system shall allow the Student to view or download their purchased QR-code ticket and allow the Club Lead to view ticket registration information for their event.	Medium	A student can view/download a valid purchased ticket. The Club Lead can view the corresponding ticket/registration information.	Allows students to access their tickets and Club Leads to monitor event registrations.
NFR-001	Non-Functional	The system shall ensure that only authenticated and authorized users can access functions appropriate to their role.	High	Students, Club Leads, and Campus Admins can access only their authorized functions. Unauthorized access is denied.	Protects event, budget, payment, and ticket information.
NFR-002	Non-Functional	The system shall respond to normal user requests within 3 seconds under expected system load.	Medium	At least 95% of requests under expected load receive a response within 3 seconds.	Ensures responsive use of the portal.

SYSTEM ACTORS

*** Club Lead (Primary Actor):**

Submits event proposals with budget requests, generates event tickets with QR codes after approval, and monitors registration records.

*** Campus Admin (Primary Actor):**

Reviews submitted proposals, evaluates proposed budgets, and records approval or rejection decisions with mandatory remarks.

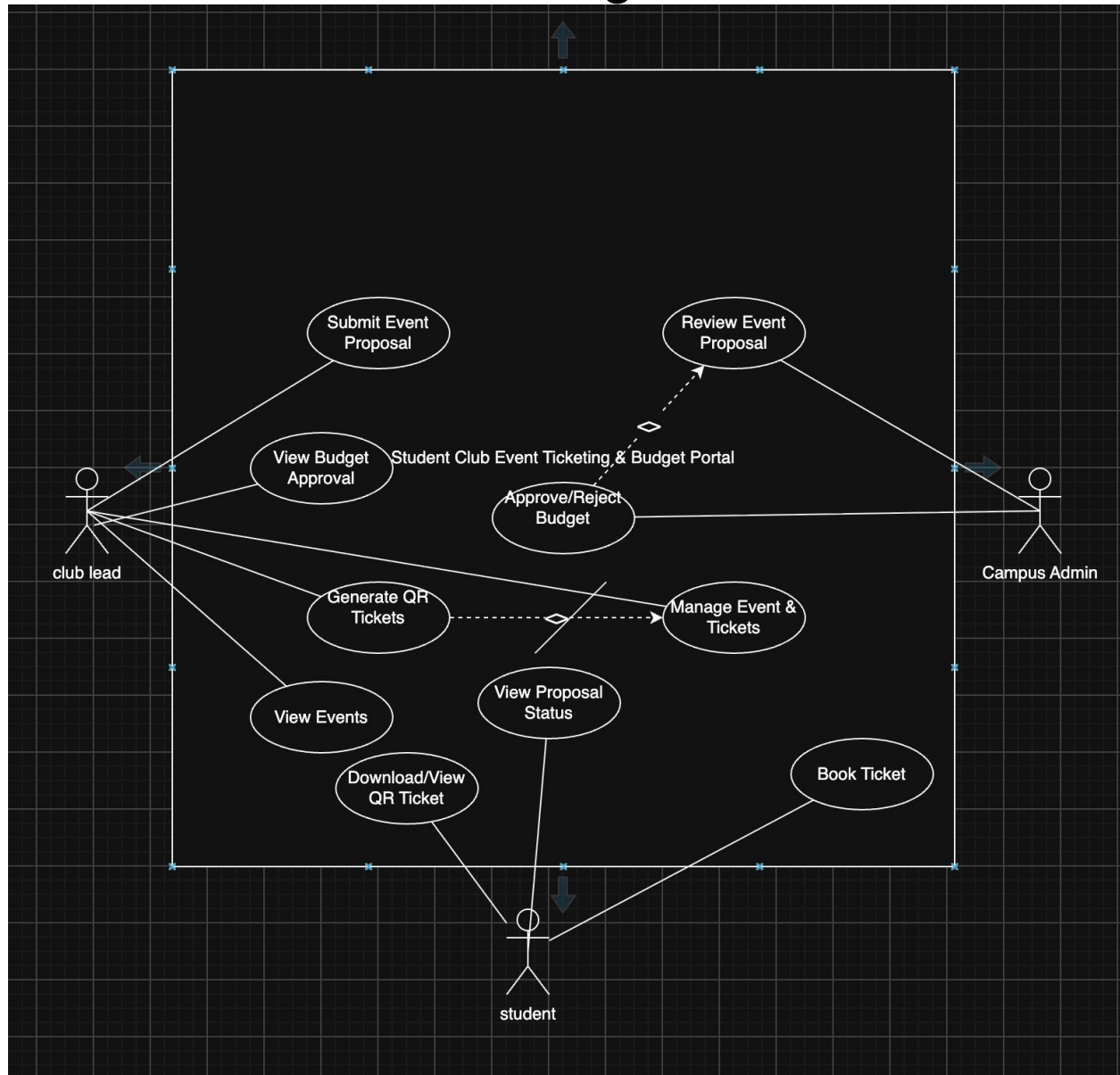
*** Student (Primary Actor):**

Browses approved events, purchases tickets via payment processing, and downloads/views issued QR-code tickets.

LIST OF SYSTEM USE CASES

- * UC-1: Submit Event Proposal & Budget
 - Primary Actor: Club Lead
 - Description: Allows the Club Lead to enter event metadata (name, date, venue, description, expected participants, budget) and submit the proposal.
- * UC-2: Review & Approve/Reject Proposal
 - Primary Actor: Campus Admin
 - Description: Enables the Campus Admin to inspect submitted proposals and record an approval or rejection decision.
- * UC-3: Generate Event QR Tickets
 - Primary Actor: Club Lead
 - Description: Allows the Club Lead to initialize and generate unique QR-code tickets for an approved event.
- * UC-4: Purchase Event Ticket
 - Primary Actor: Student
 - Description: Allows the Student to select ticket quantities, process payment, and acquire an event ticket.
- * UC-5: View & Download QR Ticket
 - Primary Actor: Student
 - Description: Enables students to access and download their issued QR-code tickets post-purchase.
- * UC-6: View Event Registration Info
 - Primary Actor: Club Lead
 - Description: Allows the Club Lead to track participant registrations and ticket availability for their events.
- * UC-7: Authenticate User
 - Primary Actor: Shared (Club Lead, Campus Admin, Student)
 - Description: Verifies credentials and enforces role-based access control.
- * UC-8: Provide Rejection Remarks
 - Primary Actor: Campus Admin
 - Description: Logs formal justification when a budget proposal is rejected.

Case Diagram



USE CASE RELATIONSHIPS («include» & «extend»)s

A. «include» Relationships

1. Purchase Event Ticket «include» Authenticate User

Rationale: Students must be authenticated to complete ticket purchases and link passes to their accounts.

2. Submit Event Proposal & Budget «include» Authenticate User

Rationale: Requires verified Club Lead authentication to prevent unauthorized submissions.

3. Review & Approve/Reject Proposal «include» authenticate User

Rationale: Requires administrative sign-in to access approval privileges.

B. «extend» Relationships

1. Provide Rejection Remarks –«extend»-> Review & Approve/Reject Proposal

Extension Point: Proposal Rejected

Rationale: Only executed when the Campus Admin rejects an event/budget proposal.

Submit Event Proposal

Primary Actor

Club Lead

Preconditions

1. The Club Lead is logged into the system.
2. The Club Lead has permission to create an event proposal.
3. The system is available.

Postconditions

Success

- Event proposal is submitted successfully.
- A unique Event ID is generated.
- Proposal status is set to **Pending Review**.

Failure

- Proposal is not submitted.
- The system displays validation errors.

Main Success Scenario

1. Club Lead selects **Submit Event Proposal**.
2. The system displays the proposal form.
3. Club Lead enters the event details and proposed budget.
4. Club Lead submits the proposal.
5. The system validates the information.
6. The system generates a unique Event ID.
7. The proposal is saved.
8. The proposal status is marked as **Pending Review**.
9. The system displays a confirmation message.

Alternate Flow – Missing Information

1. Required fields are empty.
2. The system highlights the missing fields.
3. Club Lead completes the information.
4. The process continues from Step 4.

UC-2: Review Event Proposal

Primary Actor

Campus Admin

Preconditions

1. Campus Admin is authenticated.
2. A pending event proposal exists.

Postconditions

Success

- Proposal has been reviewed.

Failure

- Proposal remains unchanged.

Main Success Scenario

1. Campus Admin opens the proposal list.
2. The system displays all pending proposals.
3. Campus Admin selects a proposal.
4. The system displays proposal details.
5. Campus Admin reviews the information.
6. Campus Admin proceeds to approve or reject the proposal.

Alternate Flow – Proposal Not Found

1. The selected proposal no longer exists.
2. The system displays an error message.
3. Campus Admin returns to the proposal list.

UC-3: Approve/Reject Budget

Primary Actor

Campus Admin

Preconditions

1. Proposal has been reviewed.
2. Campus Admin is logged in.

Postconditions

Success

- Budget decision is recorded.
- Club Lead is notified.

Failure

- Decision is not saved.

Main Success Scenario

1. Campus Admin reviews the proposed budget.
2. Campus Admin selects **Approve** or **Reject**.
3. If rejected, the system requests rejection remarks.
4. Campus Admin submits the decision.
5. The system records the decision.
6. Proposal status is updated.
7. Club Lead can view the updated status.

Alternate Flow – Proposal Rejected

1. Campus Admin rejects the proposal.
2. Campus Admin enters rejection remarks.
3. System saves the remarks.
4. Proposal status becomes **Rejected**.

UC-4: Generate QR Tickets

Primary Actor

Club Lead

Preconditions

1. Event has been approved.
2. Club Lead is logged in.

Postconditions

Success

- QR-code tickets are generated.
- Ticket inventory is updated.

Failure

- Tickets are not generated.

Main Success Scenario

1. Club Lead selects an approved event.
2. Club Lead chooses **Generate QR Tickets**.
3. The system generates unique ticket IDs.
4. The system creates QR codes.
5. Ticket records are stored.
6. Confirmation is displayed.

Alternate Flow – Event Not Approved

1. The selected event has not been approved.
2. The system prevents ticket generation.
3. An error message is displayed.

UC-5: View & Download QR Ticket

Primary Actor

Student

Preconditions

1. Student is logged in.
2. Student has purchased a ticket.

Postconditions

Success

- Ticket is displayed or downloaded.

Failure

- Ticket cannot be found.

Main Success Scenario

1. Student opens **My Tickets**.
2. System displays purchased tickets.
3. Student selects a ticket.
4. System displays the QR code.
5. Student downloads or views the ticket.

Alternate Flow – Ticket Not Available

1. Ticket record cannot be found.
2. System displays an error message.
3. Student returns to the ticket list.

UC-6: View Event Registration Information

Primary Actor

Club Lead

Preconditions

1. Club Lead is logged in.
2. The event exists.

Postconditions

Success

- Registration details are displayed.

Failure

- No registration information is available.

Main Success Scenario

1. Club Lead selects an event.
2. System retrieves registration records.
3. System displays participant list.
4. System displays ticket availability.
5. Club Lead reviews the information.

Alternate Flow – No Registrations

1. No students have registered.
2. System displays No registrations available.

UC-7: Authenticate User

Primary Actor

Club Lead / Campus Admin / Student

Preconditions

1. User has a registered account.

Postconditions

Success

- User is logged in.
- Appropriate dashboard is displayed.

Failure

- Login is denied.

Main Success Scenario

1. User enters username and password.
2. System validates the credentials.
3. System identifies the user's role.
4. Appropriate dashboard is displayed.

Alternate Flow – Invalid Credentials

1. Credentials are incorrect.
2. System displays an error message.
3. User retries login.

UC-8: Provide Rejection Remarks

Primary Actor

Campus Admin

Preconditions

1. Proposal has been rejected.

Postconditions

Success

- Rejection remarks are recorded.
- Club Lead can view the remarks.

Failure

- Remarks are not saved.

Main Success Scenario

1. Campus Admin rejects the proposal.
2. System requests rejection remarks.
3. Campus Admin enters the remarks.
4. System validates the input.
5. Remarks are saved.
6. Proposal status is updated.
7. Club Lead can view the rejection reason.

Alternate Flow – Remarks Not Entered

1. Campus Admin leaves the remarks field empty.
2. System prompts the Campus Admin to enter remarks.
3. Campus Admin enters the remarks.
4. The process continues from Step 4.

